

CDFD Runtime Paper 09: Tri-Regime Bioenergetics and Life-Number Diagnostics

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Abstract

Executive synthesis. This paper connects CDFD Runtime biology to the tri-regime Life Number Λ . The model tracks energy input, electron transport, proton coupling, stabilization load, relaxation time, and maintenance cost. The local OOL phase experiment supplies a candidate simulation boundary near $\Lambda = 1$. The correct claim is not that the runtime has solved the origin of life. The correct claim is that it provides a reproducible way to test life-like throughput thresholds under explicit model assumptions.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Biology as Bounded Throughput

In this runtime, life-like behavior is modeled as persistent organized flow. The basic local operating ratio is still

$$\Psi_s = \frac{\Phi}{C} S M_s. \quad (1)$$

For origins-of-life questions, however, a system-wide viability number is also needed because local Ψ_s can be high in one region and low elsewhere.

3 The Life Number

The Life Number is

$$\Lambda = \frac{\Phi \sigma_e \sigma_p \tau_{\text{relax}}}{S_{\text{stab}} E_{\text{maintenance}}}, \quad (2)$$

where Φ is energy input, σ_e is electron-transport effectiveness, σ_p is proton-coupling or coherence effectiveness, τ_{relax} is relaxation time, S_{stab} is stabilization or dissipation burden, and $E_{\text{maintenance}}$ is maintenance energy.

The working labels are:

- $\Lambda < 1$: decay-dominated;
- $\Lambda \approx 1$: near-critical proto-biological window;
- $\Lambda > 1$: sustained throughput in the model.

4 Runtime Implementation

The implementation lives in `engine/origins_of_life.py` and is called from `engine/biology.py`. The runtime stores channel summaries in `state.meta` keys such as `ool_C_input`, `ool_sigma_e`, `ool_sigma_p`, `ool_S_stability`, `life_number`, and `throughput_J`.

5 Experiment: OOL Phase Boundary

`experiments/notebooks/discovery_ool_phase.py` sweeps a 50 by 50 parameter grid over energy input and geochemical constraint. The report `experiments/reports/NEW_DISCOVERIES.REP` describes candidate points near $\Lambda = 1$, and `experiments/outputs/discovery_ool_phase_results.h5` stores the output.

The useful result is a model diagnostic: the runtime can locate and save a phase-boundary surface in the declared toy parameter space. The result still needs testing against laboratory chemistry, geochemical constraints, and independent origin-of-life models before any empirical claim is made.

6 Biology Discovery Scripts

The broader biology experiments include:

- `experiments/run_biology_discovery.py`: metabolic saturation, chronic drift, and tri-regime throughput;
- `experiments/run_bio_mujjabi_tests.py`: parasitic-drain and cycling-retention tests;
- `experiments/run_drug_synergy_predictions.py`: same-axis versus cross-axis intervention toy trajectories.

These are model diagnostics and hypothesis generators. They are not clinical advice or validated therapy rules.

7 Falsification Conditions

The tri-regime model fails or requires revision if:

- measured prebiotic systems show no reproducible threshold near the proposed Λ proxy;
- σ_e or σ_p cannot be measured or approximated in a domain-specific way;
- the boundary is indistinguishable from a simpler energy-only model;
- outputs become non-finite under reasonable parameter stress.

8 Conclusion

Tri-regime bioenergetics gives CDFD a bridge from local operating ratios to system-level viability. Its strength is not rhetorical certainty; its strength is that it produces explicit variables, runnable scripts, saved outputs, and failure conditions.

References

- [1] Steve Bico Mujjabi, MD. *Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant*. CDFD Part I Paper I, preprint, 2026.
- [2] Steve Bico Mujjabi, MD. *The Universal Torus Knot Hierarchy: Z_n Symmetry, the Cinquefoil Family, and the General Structure of CDFT Particle Families*. CDFD Part I Paper VI, preprint, 2026.
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- [6] Steve Bico Mujjabi, MD. *CDFD Part II: Origins of Life and Tri-Regime Bioenergetics*. Public release archive, 2026, <https://doi.org/10.5281/zenodo.20250821>.